

SYLLABUS

Semester 3



Babu Banarasi Das University
B Tech. CSE-AI

OPERATING SYSTEMS – NCS4402

Semester 4 Syllabus

MODULE 1

Introduction to Operating System and Process Concept

Operating system and functions, Classification of Operating systems, Operating System Structure, Operating System Services, System call and System program, Process concept, Process state, Process control block, Context switching, Operation on process, Threads and their management, Benefits of multithreading, Types of threads, Threading issues, CPU-scheduling, Scheduling criteria, Scheduling Algorithms, Concurrent Processes, Inter Process Communication models and Schemes

MODULE 2

Process Synchronization and Deadlock

Process synchronization, Producer/Consumer Problem, Critical Section Problem, Peterson's solution, Synchronization of hardware, Semaphore, Classical-problem of synchronization, Deadlock, Deadlock characterization, Deadlock Prevention, Deadlock Avoidance, Resource allocation graph algorithm, Banker's algorithm, Deadlock detection, Recovery from deadlock

MODULE 3

Memory Management

Memory Management, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing

MODULE 4

I/O Management and File System

File System Structure, File System Implementation, Directory Implementation and Allocation Methods, Free space Management, Kernel I/O Subsystems, Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management.

DATABASE MANAGEMENT SYSTEM – NCS4401

Semester 4 Syllabus

MODULE 1

Database System Concepts, Database Users, and Architecture

Introduction to Database System with example, Characteristics of the Database Approach, Users of Database System, Advantages and disadvantages of Using a DBMS, Implications of the Database Approach, Data Models, Schemas, and Instances, DBMS Architecture and Data Independence, Database Languages and Interfaces, The Components of Database System, Classification of Database Management Systems

MODULE 2

Data Modelling & Relational Database Management System

Data Modelling Using the Entity-Relationship Model, concepts, notation for ER diagram, mapping constraints, keys, Concepts of Super Key, candidate key, primary key, Entity Types, Entity Sets, and Attributes, Relationships, Relationship Types, Roles, and Structural Constraints, Strong vs Weak Entity Types, ER Diagrams, Naming Conventions, and Design Issues, Enhanced Entity-Relationship Modelling, Subclasses, Super classes, and Inheritance, Specialization and Generalization, Constraints and Characteristics of Specialization and Generalization, Modelling of UNION Types Using Categories, The Relational Data Model, Relational Constraints, and the Relational Algebra, Relational Model Concepts, Relational Constraints and Relational Database Schemas, Update Operations and Dealing with Constraint Violations, Basic Relational Algebra Operations, Additional Relational Operations, Examples of Queries in Relational Algebra

MODULE 3

SQL and Database Design Theory and Methodology

Structured Query Language- The Relational Database Standard, Data Definition, Constraints, and Schema Changes in SQL, Types of SQL Commands, SQL Operators and their Procedure, Insert, Delete, and Update Statements in SQL, Queries and Subqueries, Aggregate Functions, Joins, Unions, Intersection, Minus, Views (Virtual Tables) in SQL, Cursors, Triggers and PL/SQL, Functional Dependencies and Normalization for Relational Databases, Informal Design Guidelines for Relation Schemas, Functional Dependencies, Normal Forms Based on Primary Keys, General Definitions of Second and Third Normal Forms, Boyce-Codd Normal Form, Multivalued Dependencies and Fourth Normal Form, Join Dependencies and Fifth Normal Form

MODULE 4

Transaction Processing, Concurrency Control and Database Recovery

Transaction Processing Concepts, Introduction to Transaction Processing, Transaction states and State Diagram, Transaction and System Concepts, Desirable Properties of Transactions, Schedules and Recoverability, Serializability of Schedules, Concurrency Control Techniques, Locking Techniques for Concurrency Control, Concurrency Control Based on Timestamp Ordering, Multiversion Concurrency Control Techniques, Validation (Optimistic) Concurrency Control Techniques, Granularity of Data Items and Multiple Granularity Locking, Database Recovery Techniques, Recovery Concepts, Recovery Techniques Based on Deferred Update, Recovery Techniques Based on Immediate Update, Shadow Paging, The ARIES Recovery Algorithm, Database Backup and Recovery from Catastrophic Failures

COMPUTER ORGANIZATION & ARCHITECTURE – NCS 4404

Semester 4 Syllabus

MODULE 1

Computer Evolution & Arithmetic

A Brief History of computers: Von Neumann Architecture, Hardware Architecture, Computer Components, Interconnection Structures, Bus Interconnection, Register Transfer Language, Bus and Memory Transfers, Bus Architecture, Bus Arbitration Techniques, Arithmetic Logic, Shift Micro operation, Arithmetic Logic Shift Unit, Arithmetic Algorithms (Addition, subtraction, Booth Multiplication, Division).

MODULE 2

Control Unit

Control Design: Hardwired & Micro Programmed, performing of arithmetic or logical operations, Multiple Bus organization, Hardwired Control, Micro programmed control, Microinstruction, Micro program sequencing, Wide-Branch addressing, Microinstruction with Next-address field, Prefetching, Microinstruction, Pipeline control: Instruction pipelines, Pipeline performance.

MODULE 3

Processor Organization and Input-Output Organization

Processor Design: General register organization, Stack organization, Addressing mode, Instruction format, Data transfer & manipulations, Program Control, Reduced Instruction Set, Computer Memory Organization: Basic concept and hierarchy, semiconductor RAM memories, 2D & 1/2D memory organization, ROM memories, Cache memories, Concept and design issues & performance, Address mapping and replacement, Auxiliary memories: Magnetic disk, Magnetic tape and optical disks, Virtual memory, Concept implementation.

MODULE 4

Interrupts

Interrupt hardware, Types of interrupts and exceptions, Modes of Data Transfer: Programmed I/O, Interrupt initiated I/O, Direct Memory Access: I/O channels and processors, Serial Communication: Synchronous & asynchronous communication, Standard communication interfaces

CONCEPT OF ML WITH PYTHON – NAI4401

Semester 4 Syllabus

MODULE 1

Introduction to Machine Learning

Introduction, Examples of various Learning Paradigms, Perspectives and Issues, Version Spaces, Finite and Infinite Hypothesis Spaces, PAC Learning, VC Dimension.

Machine Learning using Python

Introduction to python; basic library; functions of library; implementation of library; Design, Analysis and Evaluation of Machine Learning Experiments, Other Issues: Handling imbalanced data sets.

MODULE 2

Supervised Learning Algorithms

Learning a Class from Examples, Linear, Non- linear, multi-class and multi-label classification, Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.

Advanced Supervised Learning

Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Support vector machines: Linear and Non-Linear, Kernel Functions, K- Nearest Neighbours Ensemble Learning Ensemble Learning Model Combination Schemes, Voting, Bagging: Random Forest Trees, boosting: Adaboost, Stacking.

MODULE 3

Unsupervised Learning

Introduction to clustering, Hierarchical: AGNES, DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis

Probabilistic Learning

Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesian Belief Networks, Mining, Frequent Patterns

INDIAN CONSTITUTION – NVC4401

Semester 4 Syllabus

MODULE 1

Introduction to Indian Constitution

Constitution meaning of the term - The making of the Indian Constitution - Sources and constitutional history – Philosophy of Constituent Assembly - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy. Union Government and its Administration Structure: President and Vice President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions

MODULE 2

The States and The Union Territories

State Government and its Administration: Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions – Relation between the Union and the States.

Local Administration

District's Administration Head - Role and Importance, Municipalities - Mayor and role of Elected Representative – Panchayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles - Block level Organizational Hierarchy, Village level - Role of Elected and Appointed officials - Importance of grass-root democracy